Skills Summary

Polynomial Arithmetic

Use this reference while completing your worksheet or when studying.

Skill 1 — Adding and Subtracting Polynomials

Rule: only *like* terms combine — same variable, same exponent. $ax^n \pm bx^n = (a \pm b)x^n$, and x^3 never combines with x^2.

Subtraction: rewrite $-(\cdots)$ by changing the sign of *every* term inside, then add. Answers go in standard form, highest power first.

Example: Simplify $(4x^3 - x + 2) + (2x^3 + 5x^2 - 6)$
Step 1. This is an addition, so no signs change — the whole job is sorting terms by degree and combining the matching ones.
Step 2. Cubes: $4 + 2$ gives $6x^3$. There is only one square term, $5x^2$, and only one linear term, $-x$. Constants: $2 - 6$.
$\Rightarrow$ $\boldsymbol{6x^3 + 5x^2 - x - 4}$

Try it: Simplify $(7x^2 - 3x) - (2x^2 + 4x - 5)$

check: $\boldsymbol{5x^2 - 7x + 5}$

Skill 2 — The Box Model for a Product

Model: a product is the area of a rectangle whose sides are split into the terms. One row per term of the left factor, one column per term of the top factor; each cell is one partial product, and the answer is the sum of the cells.

Count check: m terms times n terms means exactly mn cells. If you have fewer, one is missing.

Example: Multiply $(x + 5)(x^2 - 2x + 3)$
Step 1. Two terms times three terms, so the box has six cells — fill all six before collecting anything.
Step 2. From x: x^3, $-2x^2$, $3x$. From 5: $5x^2$, $-10x$, 15. Now the squares give $-2 + 5$ and the linear terms give $3 - 10$.
$\Rightarrow$ $\boldsymbol{x^3 + 3x^2 - 7x + 15}$

Try it: Multiply $(x - 3)(x^2 + 4x - 1)$

check: $\boldsymbol{x^3 + x^2 - 13x + 3}$

Skill 3 — Multiply, Then Check by Substituting

Degree check: the degree of a product is the *sum* of the degrees of the factors, and the leading coefficient is the product of the leading coefficients. Read those off first and see whether your answer matches.

Number check: pick a value of x, put it into the original factors and into your expanded answer. Two different numbers means an error.

Example: Multiply $(2x-1)(3x^2+x-4)$

Step 1. Degree 1 times degree 2 must give degree 3, with leading coefficient $2 \cdot 3$ — so expect the answer to start with $6x^3$, and use that as the first check.

Step 2. From $2x$: $6x^3$, $2x^2$, $-8x$. From -1: $-3x^2$, $-x$, 4. Collecting, the squares give $2-3$ and the linear terms give $-8-1$.

$\Rightarrow$ $\boldsymbol{6x^3 - x^2 - 9x + 4}$, and at $x = 1$ both sides give 0.

Try it: Multiply $(3x+2)(x^2-x+5)$

check: $\boldsymbol{3x^3 - x^2 + 13x + 10}$

(!) Watch out: the most common slip is a missing cell — a three-term factor times a two-term factor needs six partial products, not five. The second most common is a subtraction sign that reached only the first term inside the bracket.